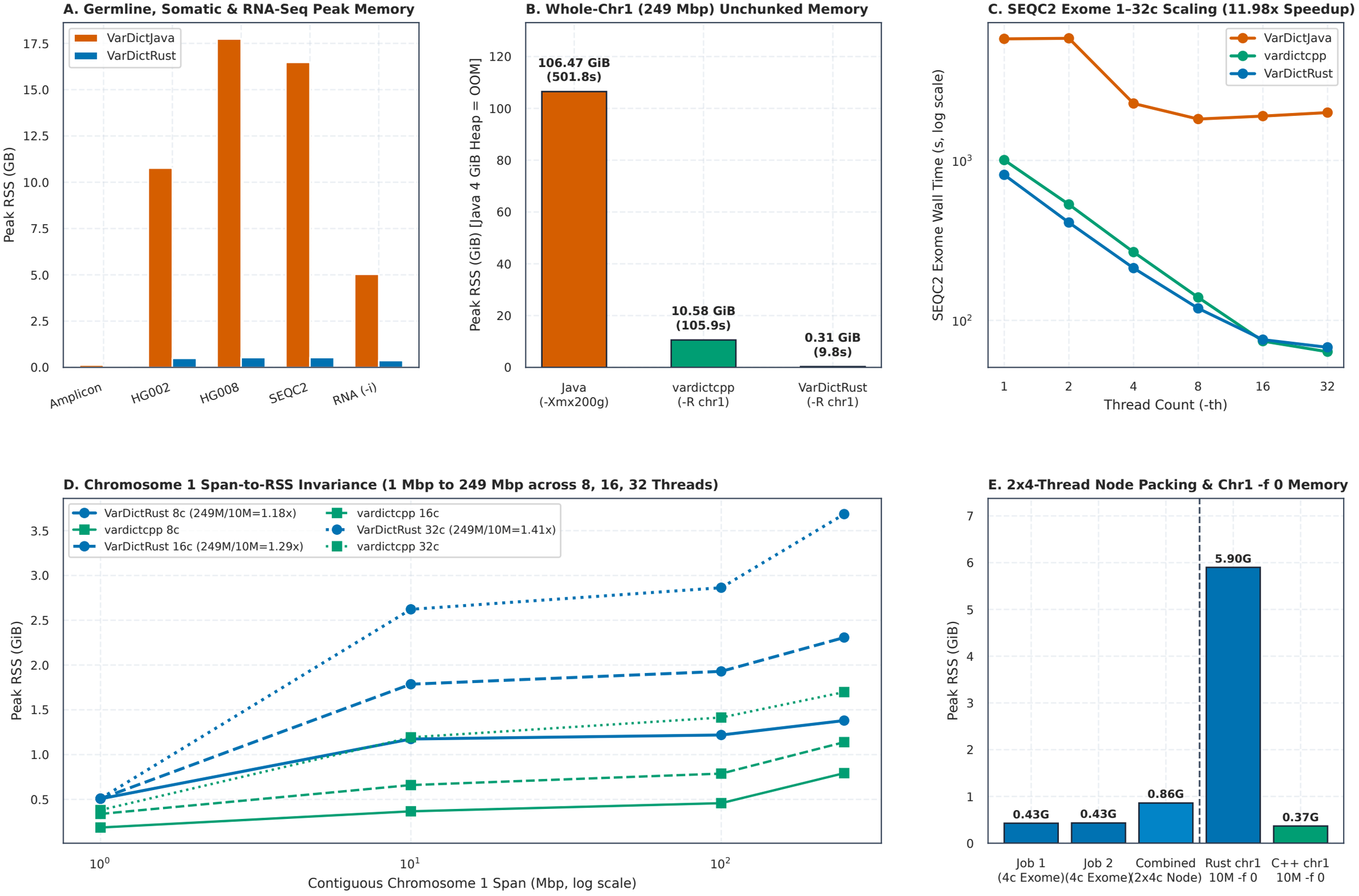
A. Germline, Somatic & RNA-Seq Peak Memory
VarDictJava
VarDictRust
Peak RSS (GB)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Amplicon
HG002
HG008
SEQC2
RNA (-i)
B. Whole-Chr1 (249 Mbp) Unchunked Memory
Peak RSS (GiB) [Java 4 GiB Heap = OOM]
0
20
40
60
80
100
120
106.47 GiB
(501.8s)
10.58 GiB
(105.9s)
0.31 GiB
(9.8s)
Java
(-Xmx200g)
vardictcpp
(-R chr1)
VarDictRust
(-R chr1)
C. SEQC2 Exome 1–32c Scaling (11.98x Speedup)
VarDictJava
vardictcpp
VarDictRust
SEQC2 Exome Wall Time (s, log scale)
10^3
10^2
1
2
4
8
16
32
Thread Count (-th)
D. Chromosome 1 Span-to-RSS Invariance (1 Mbp to 249 Mbp across 8, 16, 32 Threads)
VarDictRust 8c (249M/10M=1.18x)
vardictcpp 8c
VarDictRust 16c (249M/10M=1.29x)
vardictcpp 16c
VarDictRust 32c (249M/10M=1.41x)
vardictcpp 32c
Peak RSS (GiB)
0.5
1.0
1.5
2.0
2.5
3.0
3.5
10^0
10^1
10^2
Contiguous Chromosome 1 Span (Mbp, log scale)
E. 2x4-Thread Node Packing & Chr1 -f 0 Memory
Peak RSS (GiB)
0
1
2
3
4
5
6
7
0.43G
0.43G
0.86G
5.90G
0.37G
Job 1
(4c Exome)
Job 2
(4c Exome)
Combined
(2x4c Node)
Rust chr1
10M -f 0
C++ chr1
10M -f 0